Xiang Xiang Wang, Ph.D.

Postdoctoral Research Associate, Department of Mathematics, University of Georgia, Athens, GA 30602
xw72611@uga.edu | xiangxiangjy.github.io/xxw | 775-501-2266

Research interests: Mathematical AI and data science; geometric and topological data analysis; Grassmannian and Riemannian geometry; matrix analysis; computational mathematics; machine learning for biological and imaging data.

Academic Appointments

Postdoctoral Research Associate, Department of Mathematics, University of Georgia 2026–present
Mathematical AI for biological data: Grassmannian representations for cross-slice integration of spatial transcriptomics, and Grassmannian geometry combined with topological data analysis for viral genomics. Mentor: Guo-Wei Wei

Research Associate, Department of Mathematics, Michigan State University 2025–2026
Geometric, topological, and sheaf-based methods for single-cell and image data. Mentor: Guo-Wei Wei

Education

Ph.D. in Mathematics, University of Nevada, Reno 2025
Grassmannian geometry and quaternion matrix methods for data representation. Advisor: Tin-Yau Tam

M.Sc. in Statistics, Shanghai University, China 2020
Quaternion matrices and large-scale eigenvalue algorithms. Advisor: Qing-Wen Wang

B.Sc., North China University of Water Resources and Electric Power, China 2017
Information and Computing Science (computational mathematics)

Research Funding

Co-Principal Investigator, U.S. Department of Energy, Office of Science, Basic Energy Sciences (through Pacific Northwest National Laboratory) 2026–2029

"Self-Assembly of Tunable Molecular Memristors with Long-Range Order for Resilient and Energy-Efficient Neuromorphic Computing." $360,000, June 2026–January 2029. PI: Guo-Wei Wei. PNNL Award No. PNNL0000883191; DOE FWP 84274.

Publications

Submitted manuscripts and preprints

[P5] X.X. Wang, G.W. Wei, Mathematical AI predicts 40% and 57% influenza vaccine effectiveness against H1N1 and H3N2, respectively, for the 2026–2027 US season. *Submitted to arXiv, 2026*.

[P4] X.X. Wang, G.W. Wei, GrassTop: Grassmannian k-mer topology for viral classification and phylogenetic analysis. *Submitted to Theory in Biosciences, 2026*. arXiv:2609.13341.

[P3] X.X. Wang, S. Cottrell, G.W. Wei, Data-adaptive Grassmann manifold representations for cross-slice integration of spatial transcriptomics. *Submitted to SIAM Journal on Applied Mathematics, 2026*. arXiv:2609.12805.

[P2] S. Cottrell, X.X. Wang, G.W. Wei, A review of geometric, topological, and spectral approaches in single-cell data analysis. *Submitted to Current Opinion in Systems Biology, 2026*.

[P1] T.Y. Tam, X.X. Wang, Geometric means and their properties of Grassmannians. *Submitted to Linear and Multilinear Algebra*. arXiv:2412.15161.

Journal articles (published or accepted)

[J15] X.X. Wang, G.W. Wei, A hierarchical sheaf spectral embedding framework for single-cell RNA-seq analysis. *SIAM Journal on Life Sciences, accepted, 2026*. arXiv:2603.26858.

[J14] X.X. Wang, G.W. Wei, Multi-dimensional persistent sheaf Laplacians for image analysis. *Foundations of Data Science, 2026 (early access)*. doi:10.3934/fods.2026024.

[J13] X.X. Wang, S. Cottrell, G.W. Wei, Multiscale Grassmann manifolds for single-cell RNA-seq data analysis. *Machine Learning: Science and Technology 7 (2026) 035017*. doi:10.1088/2632-2153/ae6752.

[J12] X.X. Wang, T.Y. Tam, Color image set recognition based on quaternionic Grassmannians. *Journal of Mathematical Imaging and Vision 68 (2026) 13*. doi:10.1007/s10851-026-01289-7.

[J11] Z. Wang, D. Li, G. Wang, M. Zia, X. Wei, X.X. Wang, J. Jiang, Chaotic dynamics-regulated topological learning for patient-specific preictal state identification. *Machine Learning: Science and Technology 7 (2026) 055031*. doi:10.1088/2632-2153/aeabee.

[J10] Z.H. He, Y.Z. Xu, Q.W. Wang, X.X. Wang, Solving a system of quaternion matrix equations by using PSVD for multiple matrices with applications. *Journal of Computational and Applied Mathematics 476 (2026) 117072*. doi:10.1016/j.cam.2025.117072.

[J9] Z.H. He, T.T. Liu, X.X. Wang, Eigenvalues of quaternion tensors: properties, algorithms and applications. *Advances in Applied Clifford Algebras 35 (2025) 4*. doi:10.1007/s00006-024-01366-3.

[J8] Z.H. He, W.L. Qin, J. Tian, X.X. Wang, Y. Zhang, A new Sylvester-type quaternion matrix equation model for color image data transmission. *Computational and Applied Mathematics 43 (2024) 227*. doi:10.1007/s40314-024-02732-4.

[J7] Z.H. He, X.X. Wang, Y.F. Zhao, Eigenvalues of quaternion tensors with applications to color video processing. *Journal of Scientific Computing 94 (2023) 1*. doi:10.1007/s10915-022-02058-5.

[J6] Z.H. He, C. Navasca, X.X. Wang, Decomposition for a quaternion tensor triplet with applications. *Advances in Applied Clifford Algebras 32 (2022) 9*. doi:10.1007/s00006-021-01195-8.

[J5] Z.H. He, W.L. Qin, X.X. Wang, Some applications of a decomposition for five quaternion matrices in control system and color image processing. *Computational and Applied Mathematics 40 (2021) 205*. doi:10.1007/s40314-021-01579-3.

[J4] S.W. Yu, Z.H. He, T.C. Qi, X.X. Wang, The equivalence canonical form of five quaternion matrices with applications to imaging and Sylvester-type equations. *Journal of Computational and Applied Mathematics 393 (2021) 113494*. doi:10.1016/j.cam.2021.113494.

[J3] Z.H. He, C. Chen, X.X. Wang, A simultaneous decomposition for three quaternion tensors with applications in color video signal processing. *Analysis and Applications 19 (2020) 529–549*. doi:10.1142/s0219530520400084.

[J2] Q.W. Wang, X.X. Wang, Arnoldi method for large quaternion right eigenvalue problem. *Journal of Scientific Computing 82 (2020) 58*. doi:10.1007/s10915-020-01158-4.

[J1] G.H. Peng, X.X. Wang, Y.Z. Zhang, Multidimensional scaling analysis based on attribute reduction of bivariate mutual information. *International Mathematical Forum 12 (2017) 111–118*. doi:10.12988/imf.2017.610133.

TEACHING

Instructor of Record, University of Nevada, Reno (seven terms) 2023–2025

- Math 127 Precalculus II, lectures of more than 100 students (Winter 2023, Summer 2023, Spring 2024)
- Math 176 Calculus for Business and Social Sciences (Summer 2023, Fall 2024, Spring 2025, Summer 2025)
- Math 126E Precalculus I Expanded (Fall 2023)

Guest Lecturer, Michigan State University Fall 2025

Four lectures on manifolds and quaternionic Grassmannians in the graduate course "Mathematics-Assisted AI and AI-Inspired Mathematics"

Teaching Assistant: MTH 133 Calculus II, Michigan State (Spring 2026); Math 126 Precalculus I and Math 176, Nevada (2022–2023); grader, Theory of Positive Integers, Nevada (2021, 2023).

Training: Seminar in Teaching Mathematics and Statistics (2022); NSHE Corequisite Conference (2024).

MENTORING AND SERVICE

Mentor, Nevada GAIN Program (graduate students, including first-generation students)

Referee: Foundations of Data Science; IEEE Transactions on Signal and Information Processing over Networks; Linear and Multilinear Algebra; Electronic Journal of Linear Algebra; Numerical Algorithms; Advances in Operator Theory; Qualitative Theory of Dynamical Systems.
Reviewer: zbMATH Open; textbook review for Cambridge University Press, Mathematics for Economics and Finance, 2nd ed. (2025).
Other experience: Statistical Analyst, Office of the Provost, University of Nevada, Reno (Summer 2023): data workflows, statistical models, and dashboards for program assessment.

Research Code

Code accompanying published and submitted papers, at github.com/XiangXiangJY:

- GrassTop: Grassmannian k-mer topology for viral classification and phylogenetic analysis [P4] (Python)
- GrassST: Grassmann representations for cross-slice integration of spatial transcriptomics [P3] (Python)
- HSSE: hierarchical sheaf spectral embedding for single-cell RNA-seq analysis [J15] (Python)
- MPSL: multi-dimensional persistent sheaf Laplacians for image analysis [J14] (Python)
- MGM: multiscale Grassmann manifolds for single-cell RNA-seq data analysis [J13] (Python)
- QuaternionGrassmannian: color image sets with quaternionic Grassmannians [J12] (MATLAB)

Talks

Grassmannian methods for single-cell and spatial transcriptomics, *AMS Fall Southeastern Sectional Meeting, Kennesaw State University, Kennesaw, GA* **(45-minute talk)**	Oct 2026
Grassmannian methods for single-cell RNA-seq data, *27th Conference of the International Linear Algebra Society (ILAS 2026), Virginia*	May 2026
Grassmannian methods for visual and single-cell data, *JMM 2026, Washington, DC*	Jan 2026
Color image and video recognition based on quaternionic Grassmannians, *SIAM Great Lakes Section Annual Meeting, Chicago*	Sep 2025
Quaternion Grassmannians for color image set recognition, *JMM 2025, Seattle*	Jan 2025
Some properties of geodesic triangles in Grassmannians, *MAO 2024, Reno*	Jun 2024
Some inequalities of geometric means in Grassmannians, *JMM 2024, San Francisco*	Jan 2024
Geometric means and their properties of Grassmannians, *10th ICMAA, Kunming, China (virtual)*	Aug 2023
Geometric means and their properties of Grassmannians, *Int. Workshop on Matrix Analysis and Its Applications, Vietnam (virtual)*	Jul 2023
Arnoldi method for right eigenvalue problem of large-scale quaternion matrices, *JMM 2023, Boston*	Jan 2023
Arnoldi method for right eigenvalue problem of large-scale quaternion matrices, *JMM 2022 (virtual)*	Apr 2022

Conference Organizing and Leadership

Organizer, AMS Special Session "Mathematical and Computational Methods for Biological Structures and Data," Spring Southeastern Sectional Meeting, Birmingham, AL	Mar 2027
Co-organizer, Special Session, Joint Mathematics Meetings (JMM 2027), Chicago, IL	Jan 2027
Organizer, Minisymposium MS34 "Mathematical and Computational Methods for High-Dimensional Biological Data," SIAM Texas-Louisiana Sectional Meeting, Dallas, TX	Oct 2026
Organizer, Lecture Series in Mathematical AI and Bioscience (MATH 8970), UGA	2026–2027
Co-organizer, Special Session on Matrix Analysis and Its Applications, JMM 2026	Jan 2026
Session chair, JMM 2025 (Matrix Analysis and Its Applications) and MAO 2024	2024–2025
Local organizer, Workshop on Matrices and Operators (MAO 2024), University of Nevada, Reno	Jun 2024
Organizer, Biweekly Matrix Seminar, University of Nevada, Reno	2023–present

Honors and Awards

Love of Learning Award, Phi Kappa Phi Honor Society	2026
ILAS Travel Grant, 27th Conference of the International Linear Algebra Society	2026
College of Science Outstanding Graduate Assistant Award, University of Nevada, Reno	2025
Graduate Student Association Travel Award ($500), University of Nevada, Reno	2022
Graduate Student Association Research Grant ($1,000), University of Nevada, Reno	2021
Graduate Dean's Merit Scholarship ($10,000), University of Nevada, Reno	2021
College Graduate Excellence Award of Shanghai City	2020
Third Prize, 15th China Post-Graduate Mathematical Contest in Modeling	2018

Computing and Memberships

Computing: Python (PyTorch, Jupyter), MATLAB, R, SQL, SPSS, Git, LaTeX.
Memberships: American Mathematical Society; International Linear Algebra Society; Phi Kappa Phi.